William Du

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EDUCATION

Carnegie Mellon University — Mountain View, United States
Master of Science in Software Engineering — December 2023

Queen's University — Kingston, Canada
Bachelor of Applied Science in Computer Engineering — May 2022
Major GPA: 3.9/4.0

PROFESSIONAL EXPERIENCE

Lucid Motors — Newark, United States
Software Engineer — Feb 2024 – Present

- Improved distributed automation framework performance by parallelizing execution across 45+ TCU devices, reducing full regression runtime by 85% and accelerating release validation cycles
- Developed and maintained 800+ automated tests using Python and Robot Framework, executed 1–3 times weekly per release, ensuring continuous validation of production software across NA, EU, and KSA
- Achieved an 80% reduction in system costs by creating a cost-effective design, reducing redundancies
- Designed and implemented long-duration stability and stress testing framework of CPU, memory, thermals, disk I/O, network, and service-level metrics, enabling proactive detection of performance degradation
- Managed large-scale Linux-based test farm infrastructure using Jenkins, GitLab CI, AWS, and MongoDB, supporting automated validation workflows across multiple engineering teams

Restaurant Brands International — Miami, United States
Software Engineer Intern — Jun 2023 – Aug 2023

- Deployed feature successfully to 52% of the total device population, benefiting 2.5M unique iPhone users
- Spearheaded the development of the iOS Live Activities feature and background fetching for Burger King
- Designed a custom package and patch to allow for proper linking and copying of files into native iOS app

Queen's Predictive Geometallurgy and Geostatistics Lab — Kingston, Canada
Undergraduate Researcher — May 2021 – Aug 2021

- Increased correlation coefficient of metallurgical models by 9% to an average accuracy of 79.6% while maintaining an average accuracy of 98% in mass recovery, validated on 1,690 observations
- Explored diverse machine learning configurations, including individual models per class, category-based grouping, and holistic modeling, achieving an R-squared of up to 0.999, optimizing model prediction
- Fine-tuned model parameters across 1,690 observations, leveraging advanced OLS regression techniques

PROJECTS

Malware Classification — Aug 2023 – Dec 2023

- Developed a model to classify malware using API sequences from Catak, Oliveria, and Virus datasets
- Built malware classification model using embeddings and ensemble methods, improving accuracy by 7%
- Applied Low-Rank Adaptation and efficient fine-tuning, training only 0.06% of LlaMa's parameters

Emergency Social Network — Jun 2022 – Jan 2023

- Architected a real-time chatroom web application enabling seamless exchange of messages and images, supported by frameworks like ExpressJS and NodeJS, ensuring consistent and scalable performance
- Emphasized collaborative coding practices by integrating Continuous Integration, Development methodologies, and Test-Driven Development, ensuring both code quality and team alignment

SKILLS

Python | Java | C/C++ | SQL | Linux | Scikit-learn | TensorFlow | OpenCV | React Native | TypeScript | Swift